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‘Did I repeat so many English words?’: Stability of L1 and L2 word association responses over time and across response positions

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Abstract

The present study investigates first language (L1) and second language (L2) associative stability (AS), operationalized as the repetition rate of response words over two sessions of the same word association test. Specifically, we compare the overall patterns of L1 and L2 AS, examine between-language AS correlations within individual learners, and explore the position effect of response words on AS. A total of 40 Chinese English learners completed two sessions of continuous word association tests over two weeks, in which they were required to produce three responses to each of the 30 cue words. The results revealed (1) a significantly higher repetition rate for L2 than for L1 word association responses; (2) a significant effect of response position on repetition rate, which declined from the first to the third position in both languages; and (3) a strong correlation between L1 and L2 response repetition for individual participants. These findings provide evidence for intra-individual consistency in L1 and L2 word association behavior and highlight the discrepancies between L1 and L2 lexical organization, which are probably due to weaknesses in the organization of the L2 semantic network. Implications for vocabulary teaching and future research are also discussed.

Keywords: Associative stability; Response position; Continuous word association; Mental lexicon

1. INTRODUCTION

Studies comparing first language (L1) and second language (L2) mental lexicons through word association tests (WAT) support the assumption that there are both similarities and differences between L1 and L2 lexical organization (Fitzpatrick and Izura, 2011; Van Hell and De Groot, 1998; Zhang, 2010). Between-language comparisons have been conducted on response stereotypy and response type elicited from WAT responses (see Fitzpatrick and Thwaites, 2020 for a review), while few studies have explored L1 and L2 WAT responses in terms of associative stability (AS) over time.

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Early research adopting a test–retest approach indicated some degree of stability in L2 learners' WAT responses in terms of response number and degree of associative stereotypy (Kruse et al., 1987), but the likelihood of learners repeating WAT responses on separate occasions has not yet been explored. Meara (1982), in his seminal article, claimed that L2 learners' vocabulary is in a state of flux and that their AS should be lower than that of native speakers: that is, L2 speakers tend to give different responses when seeing the same cue words on different occasions. Other studies using repeated WATs have also documented a low degree of AS among Arabic–English (Dalrymple-Alford and Aamiry, 1970) and Dutch–English learners (Van Hell and De Groot, 1998) over time. Although it seems reasonable to posit that L2 learners are likely to vary their WAT responses in their L1 and L2, it remains unclear whether the overall degree of AS differs between L1 and L2 and whether such AS exhibits a parallel tendency toward the production of WAT responses in both languages with respect to the individual learner and individual cue word.

A related issue is whether the observed similarities and/or differences in AS elicited from the first response position can be extended to the secondary and peripheral response positions in a continuous WAT. Despite the widely held assumption that first-position responses are indicative of the strongest connections between words in the mind (Playfoot et al., 2018), researchers increasingly recognize that later responses can provide richer insight into lexical organization among children (Elbers and Van Loon-Vervoorn, 1998; Sheng et al., 2006; Spatgens and Schoonen, 2020) and adults (Lam and Sheng, 2020). These responses have been demonstrated to be associated with cue words through personal linguistic experience (De Deyne and Storms, 2008a). Additionally, they are postulated to be less stable than initial responses, which represent shared cultural knowledge within a language community. A separate analysis of WAT response stability in different positions may reveal subtle differences underlying the organization of the L1 and L2 mental lexicons. This study conducts by-participant and by-item analyses to investigate the degree of generalizability of observed between-language differences in AS and the intra-individual tendency to respond to cue words over different test sessions and different response positions.

2. LITERATURE REVIEW

2.1. Associative stability in L1 and L2 word association

In one of the earliest attempts to examine stability over time, Gekoski and Riegel (1966) sampled 33 cue words from the Michigan Restricted Association Norms (Riegel, 1965) and administered four association tests (three restricted association tests and one free association test) to 30 undergraduate students. The same set of tests was given to the participants one week later. The intra-individual repetition of free association responses between the two sessions was 43.1%, indicating that over half of the responses to the same cue word were different in the first and second tests. In a similar yet larger study by Brotsky and Linton (1967), 146 U.S. undergraduates gave WAT responses to 158 cue words on two occasions over ten weeks. The researchers calculated the percentage of repeated responses for each participant and each cue word and found that the by-participant and by-item repetition rates between sessions were both approximately 32%. In another study, Fox (1970) required 91 U.S. college students to provide WAT responses to 21 cue words and retested them after two months. He examined the AS of individual participants, as well as the AS of normative association probabilities (i.e., AS across cue words). The results showed that although individual participants repeated only 48% of the responses between test sessions, the 21 cue words elicited responses of comparable associative strength over time. This finding indicates that the AS of individual learners is lower than that of individual cue words (the normative AS). In a more recent study, Fitzpatrick et al. (2015) found that the WAT performance of younger L1 speakers was less stable than that of older L1 speakers. In their study, 36 Australian teenagers completed a WAT by responding to 100 cue words. Three months later, they performed a retest. The researchers observed that only 25% of the responses were repeated across tests, which is markedly lower than the figures reported in previous research. While there is some variation in the percentage of repeated responses across studies, previous research provides a reliable indication that the relationship between words in the mental lexicon of L1 speakers is in a state of constant change. At any given time, a different associative link dominates the associative network of a word, leading to prioritized access to and production of the associated item.

Evidence of AS in L2 learners can also be obtained from vocabulary testing research and longitudinal vocabulary development research. In a series of studies conducted to validate Lex_30, a productive vocabulary knowledge test, Fitzpatrick and Meara (2004) and Fitzpatrick and Clenton (2010) tested L2 English learners on Lex_30 twice at three-day and one-week intervals, respectively. The researchers found a significant positive correlation between the frequency profiles of WAT responses at the two test times. However, the probabilities of repeating WAT responses on the retest were only approximately 50% and 41%, respectively. In other words, while L2 learners maintained a stable WAT-responding pattern of word frequency, their actual responses varied across test occasions. Findings suggesting L2 associative instability have also emerged from longitudinal vocabulary studies. Fitzpatrick (2012) tracked a Chinese

EFL learner's one-year vocabulary development using a 30-item WAT. The researcher collected WAT response data at six distinct time points, with a seven-week interval between each collection point. A detailed examination of the learner's WAT responses at time 1 and time 2 revealed that over 60% of the responses produced at time 2 were distinct from those given at time 1. In some extreme cases (e.g., for cue words TOOTH, BOARD, CLOTH, and DIRTY), no responses were repeated in the second test. Fitzpatrick's (2012) data provide concrete evidence that individual learners' WAT responses are not stable over a short period of time and that AS is not homogeneous across cue words. In the domain of vocabulary assessment, the unstable nature of free WA has prompted some researchers (e.g., Read, 1993) to shift from productive WAT to receptive WAT with the aim of developing a more easily controlled test of vocabulary knowledge. Nevertheless, the relatively low stability of productive WAT underlines its spontaneity, thereby establishing its validity as a tool for exploring lexical organization in learners' minds.

From a theoretical standpoint, AS may be best understood from the network perspective of the lexicon. Instead of using the traditional distinction between *breadth* and *depth*, Meara (2009) proposed *size* (the number of nodes) and *organization* (the number of connections) as two fundamental dimensions of the L2 vocabulary network. This conceptualization prompts a shift in perspective, moving away from the traditional view of vocabulary as a collection of individual words and towards a network-based approach that highlights the interconnected nature of the lexicon. Focus, then, has also shifted from describing the depth of knowledge of individual lexical items to examining the impact of the structural properties of the entire network on its external performance. It has been established that the functioning of the L2 lexicon is contingent upon its size and organization. Therefore, whether a word node can be accessed stably in the lexicon is also related to these two factors. Furthermore, as the L2 lexicon is smaller and less well connected than its L1 counterpart, it can be hypothesized that differences might exist between the L1 and L2 in terms of the accessibility of associative nodes, which would emerge as a difference in AS in the present study.

2.2. Relationship between L1 and L2 word association behavior

The studies reviewed above have addressed the AS of either L1 speakers or L2 learners, which is insufficient to answer the question regarding the relationship between the L1 and L2 in individual learners' minds. Although recent years have seen a shift in focus from studying L2 learners' WAT responses against native speaker norms to comparing learners' L2 WAT responses to their L1 patterns (Fitzpatrick, 2009; Fitzpatrick and Izura, 2011; Namei, 2004), research on AS within individuals remains limited. To our knowledge, only two studies, one by Dalrymple-Alford and Aamiry (1970) and the other by Van Hell and De Groot (1998), have touched upon this issue. Dalrymple-Alford and Aamiry (1970) asked a group of Arabic–English bilinguals to complete a WAT test on two occasions, with test conditions varying according to the degree of commonality of the cue words and the language of the test. It is important to note that the researchers presented the same list of words twice in the same test, meaning that there was no time interval between the two data collection points. The results demonstrated that, on average, the bilingual participants produced 67.5% repeated responses in the Arabic WAT and 60% in the English WAT, but the difference did not reach statistical significance. Although the degree of AS in both the L1 and L2 was higher than in previous research, this discrepancy is likely attributable to the fact that participants completed the two identical tests in a single session as opposed to over an extended period. In the study by Van Hell and De Groot (1998), the researchers asked 40 Dutch–English bilinguals to complete a WAT (half in Dutch and the other half in English) and re-administered the tests to the participants two months later. The results indicated that the WAT responses in neither L1 nor L2 were as stable as previously expected. Specifically, only 42.81% of the Dutch responses and 35.78% of the English responses were repeated over two months. Overall, the findings reviewed above suggest that L2 WAT responses are less stable than L1 WAT responses, although further evidence is needed to support this conclusion. Another question to be investigated is whether the tendency to produce (un)stable WAT responses is consistent between L1 and L2 within the same bilingual individual. This concerns the convergence or divergence between L1 and L2 lexical behavior within bilingual individuals, which is a crucial line of inquiry in vocabulary studies (Channell, 2014). In light of the findings of previous individual profiling research, which indicate that bilinguals tend to demonstrate comparable frequency profiles in their L1 and L2 WAT responses (Fitzpatrick, 2009), we hypothesize that the AS in bilingual individuals' L1 will be reflected in their L2 as well.

2.3. Potential effect of response position on associative stability

Researchers have made extensive use of discrete (single) WAT in vocabulary testing and bilingual lexicon studies, with the underlying assumption that the first responses represent the strongest and most stable connections between words in the lexical network (Nelson et al., 2000; Playfoot et al., 2018). This being the case, there is an emerging recognition that later responses elicited by the continuous WAT may provide a more comprehensive understanding of the relationships between words in the human mind. One of the key questions concerning the application of the continuous

WAT is the effect of response position. While some scholars have made effective use of multiple associative responses in their inquiry (Nissen and Henriksen, 2006; Schmitt, 1998; Wolter, 2002), their focus has primarily been on collecting more extensive response data rather than on investigating the role of response position in learners' lexical organization. Some recent studies suggest that response position is a crucial factor in the production of free word associations. The continuous WAT responses of over 10,000 native Dutch speakers collected by De Deyne and Storms (2008a) demonstrated that, compared to the first response position, more situational responses, as well as fewer taxonomic responses, were produced in subsequent positions. In addition, the most frequent types and tokens of Spanish (Cabana et al., 2023) and English (De Deyne et al., 2019) associative norms have been found to change following the inclusion of responses from later positions. These findings indicate that the continuation of the association process reveals new information about how personal experience with language shapes meaningful connections in the mind. Compared to highly shared lexical and cultural knowledge, lexical representations formed through personal encounters with language should be less stable and thus may change from occasion to occasion. Evidence supporting the assumption that AS varies across response positions can also be gleaned from studies that examine associative convergence and idiosyncrasy. In a study by Lam and Sheng (2020), the researchers employed three trials of elicitation per cue to compare the convergence of WAT responses among native English speakers and L2 learners using the University of South Florida Free Association Norms (USF; Nelson et al., 2004). They found that from the first to the second trial, L2 learners produced significantly fewer USF-converging responses and that among native-speaking participants, response convergence for adjectival cues decreased significantly across trials. An additional analysis of within-group idiosyncrasy indicated that the likelihood of producing idiosyncratic responses increased as a function of trial for both groups. Given the lower AS of idiosyncratic responses compared to common responses, it can be assumed that L1 and L2 WAT responses would become less stable from the first to third response positions. However, the magnitude of such a decrease in the two languages remains to be empirically investigated. Therefore, the present study attempts to address the following three questions:

1. Are there differences between the L1 and L2 in terms of overall associative stability?
2. Are there by-participant and by-cue-word correlations between L1 and L2 associative stability?
3. Are there response position effects on L1 and L2 associative stability?

3. METHODOLOGY

3.1. Participants

Participants consisted of 40 undergraduate students recruited from three universities in Southwest and Central China (38 females, 2 males; Mean age = 19.73, SD = 1.13). Their majors included English and English education, and by the time of data collection, all the participants had passed CET-4 (College English Test Band 4), a standardized English proficiency test administered in China. CET-4 is often a requirement for graduation at many Chinese universities, and it has been shown by Jin et al.'s (2022) study that students who pass the CET-4 fall between B1 and B2 levels in CEFR (Common European Framework of Reference for Languages, Council of Europe, 2001), indicating that they can be identified as independent users of English. The participants had an average of 12.10 years of English learning, and none had any experience studying abroad. Furthermore, English was used as the language of instruction in all their undergraduate compulsory courses, and they were also required to speak English when responding to questions or completing in-class tasks, indicating their regular exposure to and use of L2.

3.2. Materials

A list of 30 English cue words and 30 Chinese translation equivalents developed by Zhang (2009) is used in the study (see full lists in Appendix A). They are high-frequency words matched on word class and degree of concreteness to avoid the confirmed effect of these lexical variables on the production of WAT responses (Nissen and Henriksen, 2006; Zhang, 2011). The word selection procedure was as follows: (1) Words from the three major classes (noun, verb, and adjective) were screened in Nation's (1990) first 1,000- and 2,000-word frequency lists; (2) derivational and inflectional words and words outside the length range of 3–10 letters were excluded; (3) the remaining words were classified as concrete or abstract based on Zhang's (2009) definitions, and a 7-point rating scale was administered to 30 non-English majors to further determine the degree of concreteness of the words. The three steps resulted in 30 words balanced on word class and degree of concreteness (see Table 1 for examples in each category). Statistical analyses showed a significant difference in the degree of concreteness between concrete and abstract words, $t = -39.550$,

Table 1
Examples of Chinese and English cue words.

Degree of concreteness	Word class	Examples of English cue words	Examples of Chinese cue words
Concrete	Noun	kitchen, dog	厨房, 狗
	Verb	smile, climb	微笑, 攀爬
	Adjective	hot, soft	热的, 柔软的
Abstract	Noun	peace, value	和平, 价值
	Verb	suppose, prefer	假设, 偏向
	Adjective	proper, essential	适当的, 必需的

$p < 0.001$, $r = 0.995$, but not among word classes, $F\ (2, 27) = 0.029$, $p = 0.971$, $\eta^2 = 0.002$. The 30 Chinese translation equivalents were selected with reference to Oxford Advanced Learner's English-Chinese Dictionary (9th Edition), Longman Contemporary English Dictionary (6th Edition), and the English-Chinese Dictionary (Chinese Edition).

3.3. Procedure

Two sessions of four continuous WATs were administered to the 40 participants over the course of one month. Session 1 consisted of two consecutive continuous WATs, one in English and one in Chinese, with a one-week interval between them. One week after the completion of session 1, the participants received session 2. To minimize the confounding effects of unrelated factors, the procedure in session 2 was identical to that of session 1, including the time of administration, the order of cue words, and the task instructions. For the convenience of data collection, we developed a web-based continuous WAT program. This program presented the 30 cue words one by one on 30 separate pages and the participants were required to type in the first three words that came to their mind. The instructions were given in written form by the first author in Chinese, the native language, to ensure that all participants were adequately informed about the procedure for completing the WAT. Additionally, no participant was informed that session 2 would be the same as session 1. To ensure the spontaneity of WA responses, no time limit was imposed on the task. However, it was required that the students complete it as quickly as possible. The program was piloted among 55 postgraduate students in advance and was refined in terms of the layout, instructions, and presentation order of cue words based on their feedback. On the first testing day, the participants received test instructions, completed a practice trial involving three words – *school* (n.), *read* (v.), and *happy* (adj.) – and then proceeded to complete the online continuous WATs individually. All the participants completed the four tasks via mobile phones.

3.4. Data analysis

As one participant dropped out during session 2, the final dataset included 156 online test papers (39 participants × 2 languages × 2 sessions). All the responses were lemmatized with reference to Zareva and Wolter's (2012) criteria: The plural forms of nouns and the third-person singular forms of verbs were transformed to their original forms (e.g., ingredients-*ingredient*, climbs-*climb*); obvious spelling errors were corrected (e.g., moutain-*mountain*); all derivations and multiword responses were retained. The percentage of repeated responses across test sessions was calculated for each participant and each cue word. By-participant Welch's ANOVA, two-way ANOVA, and post-hoc Tukey's HSD tests were conducted to compare the differences between L1 and L2 response repetition rates overall and across the first, second, and third response positions among individual learners. Additionally, a Pearson's product-moment test was performed to investigate the relationship between L1 and L2 response repetition rates within each learner. By-item analysis was also conducted on individual cue words through the same set of statistical tests. All the statistical analyses were conducted in R software (v4.1.2; R Core Team, 2021).

4. RESULTS

4.1. Differences between overall associative stability in L1 and L2

Percentages of repeated responses across two sessions of Chinese and English continuous WATs and independent-sample *t*-test results are presented in Table 2. By-participant analysis showed that, on average, the participants produced a significantly higher percentage of repeated responses in English than in Chinese WAT ($MD = 7.35$, $t = 3.044$, $df = 76$, $p < 0.01$), with a moderately large effect size (Cohen's $d = 0.703$). Similar patterns also emerged from

Table 2
Overall associative stability in L1 Chinese and L2 English.

	Chinese		English		Independent *t*-test results
	%	*SD*	%	*SD*	
By-participant	31.74	10.46	39.09	10.86	$RR_E > RR_C$**
By-item	62.68	6.76	68.55	5.37	$RR_E > RR_C$***

Note. By-participant: analysis of AS for each participant; by-item: analysis of AS for each cue word; RR_E = Repeated responses in English; RR_C = Repeated responses in Chinese.
** $p < 0.01$.
*** $p < 0.001$.

the by-item analysis: the English cue words elicited significantly more repeated responses than did the Chinese cue words ($MD = 5.87$, $t = 3.724$, $df = 58$, $p < 0.001$), with a large effect size (Cohen's $d = 0.868$). Moreover, a general advantage was observed for AS across cue words compared to learners, suggesting greater stability for normative associative responses than individual WAT responses.

4.2. Relationship between L1 and L2 overall associative stability

Pearson's product-moment test was conducted on the percentages of repeated L1 and L2 responses for each learner and each cue word. It revealed a significant positive correlation between the two languages, $r = 0.636$, $p < 0.001$, 95% CI [0.400, 0.792] for participants, which indicates a parallel tendency for individual learners to produce (un)stable WAT responses in L1 and L2. However, no significant correlation was found between L1 and L2 AS for cue words, $r = 0.1547$, $p = 0.414$, 95% CI [−0.2177, 0.4878], which suggests that even though two words are equivalent in meaning, they do not reliably elicit WAT responses from the learner group with comparable degrees of stability.

4.3. Effect of response position on L1 and L2 associative stability

Percentages of repeated responses among the three response positions and between L1 and L2 are shown in Table 3. For both L1 and L2 and on the level of both participant and cue word, there was a general declining trend in AS from the first to third response position. Furthermore, by-item AS was consistently higher than by-participant AS for every response position.

Differences in AS among first, second, and third response positions as well as between L1 and L2 were analyzed across participants. One of the six data sets was not normally distributed, and Levene's Test for Homogeneity of Variance revealed significant differences in variances across groups, $F(5, 228) = 5.117$, $p < 0.001$, indicating that a two-way Welch's ANOVA would be more appropriate (see results in Table 4). The test revealed a significant main effect of response position, $p < 0.001$, with a large effect size, $\eta^2 = 0.663$, whereas the main effect of language did not reach statistical significance, $p = 0.156$, $\eta^2 = 0.002$. The interaction effect between language and position was also non-significant, $p = 0.621$, $\eta^2 = 0.0014$.

Post hoc Tukey's HSD tests further identified significant differences between the first and second ($MD = 22.645$, 95% CI [18.6892, 26.6092], $p < 0.001$), the first and third ($MD = 29.914$, 95% CI [25.9538, 33.8738], $p < 0.001$), and between the second and third positions ($MD = 7.265$, 95% CI [3.3046, 11.2246], $p < 0.01$) for AS in Chinese. Similarly, significant differences were found between the first and second ($MD = 21.112$, 95% CI [17.1515, 25.072], $p < 0.001$), the first and third ($MD = 27.180$, 95% CI [23.2200, 31.1400], $p < 0.001$), and between the second and third positions ($MD = 6.069$, 95% CI [2.1085, 10.0285], $p < 0.05$) for AS in English.

Differences in AS among the three response positions and between the two languages were also investigated across cue words. All data sets were found to be normally distributed and no significant difference was revealed in variance among the groups, $F(5, 174) = 1.118$, $p = 0.353$, meaning that a two-way Analysis of Variance (ANOVA) could be conducted (see results in Table 5). The ANOVA test revealed a significant main effect of response position on AS, with a large effect size, $p < 0.001$, $\eta^2 = 0.625$. Neither the main effect of language nor the interaction between response position and language was significant, $p = 0.412$, $\eta^2 = 0.004$; $p = 0.442$, $\eta^2 = 0.009$.

Post hoc Tukey's HSD tests revealed significant differences between the first and second ($MD = 21.195$, 95% CI [15.7989, 26.5906], $p < 0.01$), the first and third ($MD = 32.904$, 95% CI [27.5081, 38.2998], $p < 0.01$), and between the second and third positions ($MD = 11.709$, 95% CI [6.3134, 17.1051], $p < 0.001$) for AS in Chinese. Similarly, differences were also significant between the first and second ($MD = 17.094$, 95% CI [11.6979, 22.4896], $p < 0.001$), the first

Table 3
L1 and L2 associative stability in the first, second, and third response positions.

	Response Position	Chinese		English	
		%	*SD*	%	*SD*
By-participant	First	33.50	12.07	38.72	10.28
	Second	12.73	7.87	15.73	6.96
	Third	6.67	6.58	8.46	5.79
By-item	First	64.96	11.40	70.34	9.00
	Second	44.27	10.21	52.74	8.90
	Third	31.37	9.02	37.69	11.26

Table 4
Results of two-way Welch's ANOVA analysis across participants.

Source	Sum of squares	*df*	*F*	*p*	$\eta 2$
Language	155	1	2.024	0.156	0.006
Position	18,988	2	123.870	0.001	0.666
Language × Position	73	2	0.478	0.621	0.004
Residuals	17,475	228			

Table 5
Results of two-way ANOVA analysis across cue words.

Source	Sum of squares	*df*	Mean Square	*F*	*p*	$\eta 2$
Language	77	1	77	0.675	0.412	0.004
Position	33,172	2	16,586	144.903	0.001	0.625
Language × Position	188	2	94	0.820	0.442	0.009
Residuals	19,917	174	114			

and third (MD = 33.333, 95% CI [27.9372, 38.7289], p < 0.001), and between the second and third positions (MD = 16.239, 95% CI [10.8435, 21.6352], p < 0.001) in terms of English AS.

5. DISCUSSION

The analyses yielded four important findings. Firstly, L2 WAT responses demonstrated greater stability than L1 WAT responses for both participants and cue words over a two-week test period. Secondly, the tendency for individual learners to produce (un)stable L1 responses was mirrored in their L2 performance. Thirdly, there was a decline in AS from the first to third response positions for both L1 and L2. Finally, in comparison to individual learners' associative behavior, WAT responses for cue words, which represent group associative behavior, were more resistant to change. Overall, the findings suggest distinct patterns in the workings of the L1 and L2 mental lexicons at the group level, which contributes to the ongoing discussion concerning the relationship between the L1 and L2 lexicons (Cangır and Durrant, 2021). In the following section, we discuss these findings in relation to previous research and existing theoretical models of the mental lexicon and the development of lexical representations.

5.1. Differences between L1 and L2 lexical organization

Overall, the significant differences found between L1 and L2 associative stability not only align with the findings from previous comparisons of L2 learners' WA behavior to their L1 behavior or native speaker norms (Fitzpatrick and Izura, 2011; Henriksen, 2008; Zareva, 2007) but also provide corroborating evidence for the network perspective on the structural properties of the L1 and L2 mental lexicon (Feng and Liu, 2024; Meara, 2009; Wilks and Meara, 2002). The former strand demonstrates that the L2 mental lexicon is at least quantitatively different from the L1 lexicon in terms of the number of associative links (Zareva, 2007), the amount of time needed to access these links (Fitzpatrick and Izura, 2011), and the number of high- and low-frequency words stored in the lexicon (Henriksen, 2008), while the latter shows that the

L1 lexicon is both denser (Meara, 2009) and more modularly structured (Feng and Liu, 2024) than the L2 lexicon. The current findings add to the existing literature by revealing that the probability of accessing the same associative responses differs across the L1 and L2 lexicons.

The finding that L2 WAT responses were more stable than L1 WAT responses over time seems contradictory to the findings reported by Dalrymple-Alford and Aamiry (1970) and Van Hell and De Groot (1998). Outside the domain of stability research, there is also abundant evidence in the broad WAT literature supporting the advantage of L1 over L2 in various aspects. For example, the L1 lexicon demonstrates more semantic associations between words compared to the L2 lexicon (Zhang, 2010). In addition, bilinguals produce L1 WAT responses not only in a larger number (Zareva, 2010) but also more quickly (Fitzpatrick and Izura, 2011; Van Hell and De Groot, 1998). Other researchers, adopting the word-naming task, have also demonstrated the relative instability of the L2 compared to the L1 in response time measures (e.g., de Bot and Lowie, 2010). To seek possible explanations for the reverse pattern found in this study, it is necessary to re-examine what stability really measures. In our operationalization, stability refers to the repetition rate of WAT responses on the second presentation of cue words. It can reasonably be assumed that stability primarily depends on the associative domain (i.e., vocabulary size) of the learner because a larger vocabulary is the precondition for a greater diversity of potential responses in WAT performance. This is consistent, to some extent, with Wolter's (2001) suggestion that when comparing the WAT responses of L2 learners to those of native speakers, it is important to consider that "native speakers at their disposal have a wider range of possible responses" (p. 65). This reminder remains valid when comparing the L1 and L2 within the same individual. Unlike simultaneous bilinguals, the participants in this study had already established their L1 lexical network before learning the L2, which results in a difference between L1 and L2 vocabulary sizes. The higher degree of AS in L2 WA can be attributed to the relatively limited capacity to provide different types of words on the two occasions of the test. Although Van Hell and De Groot (1998) reported an L1 advantage in AS over L2, this finding could have been biased by their scoring method. While our approach treated two response words as identical only when their lemmatized forms were the same, they considered a cue–response pair as identical even when it bore a synonymous relationship. Category-based WAT studies indicate that L1 speakers tend to produce a greater number of synonymous responses than L2 learners in the WAT (Fitzpatrick, 2006; Fitzpatrick and Izura, 2011), which may explain why the L1 Dutch WAT responses were found to be more stable than the L2 English WAT responses in Van Hell and De Groot's (1998) study.

Another plausible explanation concerns not only vocabulary size but also the organization of the L1 and L2 mental lexicons. Given that the continuous WATs used only high-frequency words and that the participants were mostly English majors, it would not be difficult for them to provide six different responses to each cue word in the two test sessions. If this were the case, the source of the between-language discrepancy would then lie in L2 lexical organization: that is, in the lack of connections between words and/or the relative weakness of these connections. These two aspects are assumed to be the source of *fuzziness* in the ontogenesis model (OM) of L2 lexical representation (Bordag et al., 2021). According to the OM, the central property of L2 lexical representations is fuzziness – the ambiguous and inexact encoding of lexical representation across three dimensions (linguistic domain, mapping, and network) and among three components (phonology, orthography, and semantics). While the WA process involves all dimensions and components of lexical knowledge, the most pertinent to AS seems to be the network dimension of L2 semantic representations. Given the close relationship between the frequency of word usage and the development of lexical representations (Jefferies et al., 2006), there would be an L1 advantage over the L2 in both the number and stability of representations. The sufficient and robust semantic connections in the L1 lexicon imply that L1 WA would entail competition among numerous semantic representations, with a low probability of accessing the same representation at different points in time. In contrast, L2 words would activate fewer and weaker semantic representations, and, at the same time, representations that were established earlier would become more prominent and be more likely to be accessed and produced in the test. These inferences lead to the conclusion that AS could serve as a useful indicator of the accessibility of semantic representations in the mental lexicon over a given period. Greater AS corresponds to a lower likelihood of accessing different representational nodes linked to the node of the cue word in the semantic network, while lower AS is a manifestation of numerous representational nodes existing in the network that are readily accessible at any time.

To provide further illustration, one participant (labeled H) was asked to complete a third session of L1 and L2 WATs one month later, and this time only three cue words – *drink*, *smile*, and *hot* – were used. She was given 30 s to generate as many responses as possible to each cue. Appendix B presents all the responses she produced in the three WAT sessions. In the first two sessions, H produced entirely different responses to the three cue words in the L1 test but repeated over half of the responses in L2. In the third session, all of the responses that had been given in the first two sessions of the L2 WAT reappeared. In contrast, only one response – 冷的 (cold) to the cue 热的 (HOT) – occurred in all three sessions of the L1 WAT. In a brief interview, H indicated that she did not even realize that she had repeated so many English words in the three test sessions. This evidence lends further support to the hypothesis that the increased AS of L2 WAT responses is a consequence of the limited number and strength of semantic representations in the L2 lexicon.

The distinction between L1 and L2 AS is also in line with the findings of research adopting a network approach to investigate the L1 and L2 mental lexicons. Stella et al. (2017) modeled the mental lexicon as a multi-layered network comprising various relationships, such as free association, co-occurrence, similarity of features, and phonology. In a similar vein, Kovacs et al. (2021) found that the connections in the Hungarian WA network are multi-layered, comprising semantic, syntactic, phonological, and encyclopedic associations at the same time. The findings of both studies highlight the complexity of the L1 lexicon's structure, suggesting that the outward-extending associative pathways of an L1 lexical item may be highly diverse and, consequently, less predictable. In the L2 lexicon, however, fewer words possess multiple layers of associative paths, leading to straightforward access to the most prominent associative responses. A similar distinction can be observed in studies employing a receptive approach to the WAT. Wilks and Meara (2002) asked a group of French learners to identify possible associative links among 40 sets of words and compared their performance to that of native French speakers. The results demonstrated that the L1 association network was more densely organized than the L2 network. Meara (2009) interpreted this finding as an indication of the smaller size and simpler links in the L2 lexicon, which may be the source of stability in L2 WA behavior in the present study. The L1–L2 structural difference is further supported by Feng and Liu (2024), who conducted a complex network analysis of Chinese students' semantic fluency data (derived from the *fruits and vegetables* category). They found that the degree of modularity in the L1 lexical network was higher than that in the L2 network, which illustrates the L1 network's superior capacity to form modules or subgroups among peripheral words based on similar features. It could be postulated that L1 speakers form associations not only in a less stable way but also with less frequent items than their L2 counterparts.

It should be acknowledged that, in addition to the aforementioned explanations, there are other sources of influence contributing to the between-language differences, particularly those stemming from the linguistic properties of the cue words. For example, English words often have multiple meanings or senses (Srinivasan and Rabagliati, 2015), which pose challenges for L2 learners. To illustrate, VALUE can be used to talk about (1) how much something is worth in money, (2) the quality of being useful or important, and (3) beliefs about what is right and wrong in life. This polysemic property may lead to a concentration of associative responses to a relatively familiar sense of a particular L2 cue word rather than a distribution of diverse responses to various senses of L1 cue words. Our response data offer preliminary support for this assumption. Participants produced a large proportion of responses to the first sense of VALUE, such as *price*, *money*, and *jewelry*. In addition to polysemy, the degree of abstractness of words (Williams, 1994) may also play a role in the stability of lexical organization, which needs to be further examined in future research.

With regard to cue words, our results showed that in both L1 and L2, the normative associative responses derived from each cue word were more stable than individual WAT responses, which is consistent with previous research examining normative associative behavior over time. Fitzpatrick et al. (2015) used associative stereotypy to test the reliability of WA norms obtained from English-speaking teenagers over three months and found that stereotypy scores between the two test times were highly correlated, even though individual subjects produced an average of 25% repeated responses over time. The difference between individual and group WAT behavior is also consistent with Palmberg's (1987) longitudinal study of vocabulary development. He asked a group of EFL learners to produce words beginning with the letter M or R spontaneously and tested them on each letter five times over a period of five months. Although the L2 children demonstrated a clear upward trend in the number of words produced over time, the by-participant analysis revealed that this trend was obvious for only one learner. Based on these lines of evidence, we could assume that the variation in individual learners' lexical behavior would diminish when the group pattern is the focus of investigation, which further demonstrates the reliability of applying WA norms to the study of semantic connections within different populations. Among the normative responses to a cue word, the most frequently produced response is often referred to as the *primary response* (Meara, 1978). This type of response reflects the shared cultural knowledge that forms the foundation for communication among members of the same language community (Deese, 1966). In contrast, individual participants' WAT responses are closely related to their own unique experience with language and are thus prone to change over time. The notion of such a difference is also reflected in the relative strength hypothesis (Nelson et al., 2000), according to which recent experience is the source of intra-individual variation, whereas life experience is the source of inter-individual variation. Changes in the responses may be attributed to specific recent life events that are related to the concept denoted by the cue word (e.g., exposure to video clips or images of puppies may lead to an increased likelihood of producing *cute* as the response to DOG, rather than producing its coordinate *cat*). Associative norms based on responses produced by all the group members are less likely to be affected by experiential recency and are more indicative of the experiential commonality of the group.

5.2. Similarities between L1 and L2 lexical organization

The positive correlation found between L1 and L2 AS for individual learners is consistent with the findings of studies adopting an individual profiling approach to examining bilingual WAT responses. These studies demonstrate that individual preferences for WAT response types remain stable across cue word lists (Fitzpatrick, 2007; Higginbotham, 2010) and that response preferences in L1 WAT are also reflected in L2 WAT (Fitzpatrick, 2009). Studies of Chinese–English bilingual children's WAT patterns also present evidence of a strong correlation between L1 and L2 preferences for paradigmatic responses, especially when cue words are nouns (Sheng et al., 2006). The present study provides further evidence of the similarity between L1 and L2 associative behavior within a single learner, which could have two possible explanations. One is that individual response patterns could be the result of certain types of response strategies when performing the WAT, which has also been discussed previously by some scholars (e.g., Fitzpatrick, 2009). It is likely that the participants in this study consciously altered their responses when seeing the cue words a second time and that the conscious implementation of such a strategy was transferred from L1 to L2. The other explanation is related to the organizational similarity between the L1 and L2 lexicons. The formation and restructuring of the L2 lexical network occurs when a sophisticated and well-developed L1 network is in place (Wolter, 2006). Thus, it is likely that learners who have built many equally strong lexical connections in the L1 network would have formed the same kind of connections for individual lexical items in the L2 network. Similarly, learners whose L1 network demonstrates varying degrees of connective strength between words would also develop an L2 network characterized by a predominance of a small number of words possessing robust connections with each other, which ultimately leads to the reappearance of these words in different WAT rounds. Whatever the underlying reason may be, our findings support the close association between the well-established L1 lexicon and later-acquired L2 lexicon in a single learner's mind, contributing to the wider discussion on bilingual semantic/conceptual representations, especially for L1–L2 translation equivalents (Ameel et al., 2009; Francis, 2005; Kroll, 1993) and the intra-individual convergence of L1 and L2 development in different bilingual populations (Brown, 2015; Derwing et al., 2009).

The observed relationship between learners' two languages in terms of AS has two implications. First, since there is a low degree of AS in WAT performance irrespective of language, future research should also pay attention to what L2 learners produce in the WAT, and especially to the production of different responses across test sessions in longitudinal studies, rather than focusing solely on response type, native-likeness, and other patterns derived from the response data. The latter approach dominates the existing WAT literature and offers valuable insight into the development of learners' depth of vocabulary knowledge over time (Dóczi and Kormos, 2015; Fitzpatrick, 2012; Schmitt, 1998). However, the former approach could aid in capturing the restructuring process of learners' mental lexicons, which is particularly insightful when intra-individual variation is of concern. Second, there is a need for more research on the relationship between L1 and L2 lexical networks on the developmental trajectories of L1 and L2 lexical networks. As the present study indicates, L1 and L2 AS are comparable over a short period of time, and it is necessary to explore what other features of the L1 and L2 mental lexicons develop in parallel with each other and under what conditions one language lags behind the other in terms of lexical development.

5.3. The role of response position in L1 and L2 associative stability

This study also revealed a significant decrease in AS from the first to the third response position for L1 and L2, which aligns with previous findings regarding the effect of response position on learners' WAT behavior (De Deyne and Storms, 2008a, 2008b; Lam and Sheng, 2020; Sheng et al., 2006). An average participant repeated surprisingly few responses – about two responses in L1 and three responses in L2 in the third response position. The declining AS aligns with Sheng et al.'s (2006) finding that from the first to the third response position, bilingual children produced a lower percentage of paradigmatic responses and a higher percentage of syntagmatic responses. Paradigmatic responses reflect semantic relationships such as synonymy, antonymy, and coordination and are probably more stable than syntagmatic responses, which are formed primarily through the co-occurrence of words in sentence contexts and are dependent on learners' individual language experiences. This declining trend also parallels the changes in semantic and type–token patterns of WAT responses across response positions. In the second and third positions, more situational responses occurred, which reflect individual, unique language experiences and are thus generally less stable compared to the taxonomic responses that were prominent and highly common in the first position (De Deyne and Storms, 2008a). Moreover, the second and third positions do not only constitute more tokens of responses but also add more types of responses to the association data, reflecting the productivity of human language (De Deyne et al., 2019; De Deyne and Storms, 2008b). From a theoretical perspective, the confirmed effect of response position supports the local processing assumptions within the spreading activation theory (Collins and Loftus, 1975). Upon stimulation of a concept, activation spreads out along the pathways of the conceptual network with declining magnitude, which corresponds to the declining AS. As the associative procedure continues, activation spreads to weaker links in the network, resulting in the activation and retrieval of less stable responses.

The effect of language, however, was found to be non-significant when the effect of response position was accounted for. This partly corresponds to the findings in Lam and Sheng's (2020) study. They used repeated WAT to investigate the within-group idiosyncrasy of responses among Chinese–English bilingual adults as well as native English speakers. The results showed a significant main effect of the elicitation trial. In other words, during the production of the third response, both groups of participants produced words that were unique to themselves. In comparison, the main effect of language was only marginally significant ($p = 0.04$). A closer look at the data reveals that in the first position, the native speaker group and the L2 learner group produced approximately 23.2% and 27.77% idiosyncratic responses, respectively, with a difference of 4.57%. However, the difference in the percentage of idiosyncratic responses between the first and second positions was 19.27% for native speakers and 20.33% for L2 learners. Our findings suggest that the effect of response position may overshadow the effect of language on WAT response patterns, not only in terms of response idiosyncrasy but also in terms of stability. It further suggests that multiple test- and learner-related factors interact with one another to affect the production of WAT responses, but they differ in their degree of importance and magnitude of their effects.

We need to point out that interpretations of the intra-individual AS may differ substantially between studies. Researchers (e.g., Nelson et al., 2000) hoping to establish free association norms and apply reliable cue–response relationships to other verbal behavior studies should avoid unstable responses from the second and third positions. However, in research on the L2 and bilingual mental lexicon, less stable responses may play an important role in revealing details of learners' lexical organization. Recent neurocognitive research using receptive WAT and fMRI technology has demonstrated that idiosyncratic responses tend to activate the semantic control network, whereas stereotypical responses – which often appear in the first position – are linked to the engagement of episodic memory (Krieger-Redwood et al., 2023). Thus, the less stable and more idiosyncratic responses elicited in later positions may reveal crucial information about verbal creativity in learners' L1 as well as L2.

6. CONCLUSION

The present study used continuous word association tests to explore the stability of Chinese EFL learners' L1 and L2 word association responses over two weeks. The findings showed that (1) L2 word association stability was greater than L1 word association stability over the period; (2) there was a tendency for individual learners to produce (un)stable word association responses consistently in L1 and L2; and (3) stability decreased significantly from the first to third response position in L1 and L2. We demonstrated the relevance of word association stability to the concept of fuzziness in lexical representation and discussed how the greater stability in L2 word association responses reflects the problems in the organization of the L2 semantic network.

The present study has several limitations. First, we adopted only a two-week interval to investigate associative stability over time. Although previous research has indicated that associative stability is not affected by the length of the test–retest interval, whether such a conclusion holds true for L2 associative stability remains to be investigated. Second, the stability patterns for groups and individuals may not be generalizable to WAT responses to low-frequency words, which needs to be examined in future research. Finally, the present study does not address the relationship between associative stability and other extensively investigated aspects of WAT responses. For example, category-based analysis of stable and unstable WAT responses may provide valuable insights into what types of semantic connections can be more easily integrated into the mental lexicon. These aspects warrant future investigation by researchers interested in lexical organization in the L1 and L2 mental lexicons.

DECLARATION OF CONFLICTING INTEREST

The authors declare no conflicting interest.

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Data availability

Data will be made available on request.

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APPENDIX A. ENGLISH AND CHINESE CUE WORDS

Degree of concreteness	Word class	English cue words	Chinese cue words
Concrete	Noun	Kitchen	厨房
		Dog	狗
		River	河流
		Tree	树
		Computer	计算机
	Verb	Smile	微笑
		Climb	攀爬
		Drink	喝
		Throw	扔
		Sleep	睡
	Adjective	Hot	热的
		Soft	柔软的
		White	白色的
		Old	老的
		Dry	干的
Abstract	Noun	Peace	和平
		Value	价值
		Faith	信念
		Love	爱
		Attempt	尝试
	Verb	Suppose	假设
		Prefer	偏向
		Imply	暗指
		Attract	吸引
		Realize	实现
	Adjective	Proper	适当的
		Essential	必需的
		Complex	复杂的
		Specific	特定的
		Mental	心智的

APPENDIX B. RESPONSES FROM PARTICIPANT H IN THREE SESSIONS

Language	Cue	Session	Responses
Chinese	喝	1	水, 饮料, 奶茶
		2	中药, 咖啡, 牛奶
		3	咖啡, 水, 冰, 药, 牛奶, 汽水, 椰汁, 甜, 味道
	微笑	1	大笑, 嘲笑, 冷笑
		2	笑容, 美丽, 友好
		3	友好, 面容, 悲伤, 优雅, 奔跑, 流泪, 笑傲江湖
	热的	1	冷的, 辣的, 空调
		2	冷的, 咖啡, 温度
		3	冷的, 食物, 面, 热腾腾, 汤, 热水, 温泉, 天气
English	Drink	1	milk, eat, water
		2	eat, wine, water
		3	water, milk, hot, white, eat, food, flavor, cup, tea
	Smile	1	happy, laugh, kind
		2	happy, sad, laugh
		3	happy, laugh, sadness, mood, face, unhappy
	hot	1	cold, dance, weather
		2	cold, dog, weather
		3	cold, summer, weather, water

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